# 一种基于新型Salomon耦合映射格的彩色图像加密技术

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**摘要**

本文提出了一种基于新型Salomon耦合映射格的彩色图像加密技术。该方法利用Salomon耦合映射格的混沌特性，实现了对彩色图像的高效加密。通过对图像的像素值进行扰动和置换，增强了图像的安全性和抗攻击能力。实验结果表明，该加密方法在保持图像质量的同时，能够有效抵御常见的攻击手段，如统计分析攻击和差分攻击。

**关键词：**图像加密; SCML; 混沌系统; 动态路径扩散

## 1 引言

由于混沌系统具有初值敏感性、伪随机性与轨道不可预测性，其动力学特征与密码学所要求的混淆和扩散相吻合，因而被广泛引入图像加密 [1,2]。与低维或一般高维混沌映射相比，时空混沌在时间与空间两个维度上同时呈现混沌，能够提供更宽的参数区间和更丰富的伪随机序列 [3]。Kaneko于1989年提出的耦合映射格子（Coupled Map Lattice, CML）是该类模型的经典形式：它以离散时间、离散空间、连续状态的格点阵列为载体，通过近邻扩散耦合将局部映射的混沌行为在空间上传播 [4]。此后，一类工作沿相邻或全局耦合方向扩展参数范围：Khellat等提出全局非局部耦合格子（GNCML） [5]，Rhouma等将单向耦合格子（OCML）用于彩色图像加密 [6]，Wang与Liu则构造了具有宽参数域的两参数混合耦合格子（TWMCML，亦称全混沌耦合格子）并用于隐私图像加密 [7]。相对标准CML，这些模型在更宽参数区间内表现出更强的混沌行为，但其耦合仍主要局限于局部邻域，系统整体动力学难以摆脱近邻约束。另一类工作改用非邻耦合以打破空间局域性：Zhang与Wang提出混合线性–非线性耦合Logistic映像格子（MLNCML），通过在耦合项中同时引入线性与非线性分量扩大混沌参数区间并抑制空间同步，并据此构建对称图像加密算法 [8,9]；随后又以非邻耦合格子（NCML）使扰动能够跨格点快速扩散 [10]。Wang等进一步给出非邻动态耦合格子 [11]与多重耦合格子（MCML） [12]，Tao等提出多重动态耦合格子（MDCML） [13]。与此并行，动态耦合系数模型（LDCML） [14]、交叉耦合格子（CCML） [15]、多重动态耦合系数格子 [16]、分段多动态耦合系数格子 [17]以及二维MLNCML [18]相继被提出，其共同思路是令耦合强度随时间或空间变化，或替换、升维局部映射，从而削弱固定耦合参数带来的可预测性。上述改进分别缓解了局部耦合过窄或参数空间受限的问题，但耦合方式、耦合系数与格点映射往往难以同时优化。近年工作仍围绕这三项结构要素展开：Du等提出非邻动态耦合格子NCMLP，以扩大参数范围并增强序列复杂性 [19]；Li等将非邻CML用于多人脸图像加密 [20]；Ding等构造指数余弦多项式非线性耦合与动态非局部相互作用的时空系统 [21]；Wang等给出正弦–二次映射非邻CML（SQMCML）以及由元胞自动机引导的正弦–余弦优化CML（SCOCML） [22, 23]；

Liu等则将模型推广至三维CML [24]。由此可见，耦合模式、耦合系数与格点映射模型仍显著影响时空混沌的强度与稳定性，三者的协同设计仍是将CML用于安全序列生成的核心问题。

DNA分子具有高存储密度、天然并行性与低能耗等特点，其碱基互补与运算规则为图像加密提供了不同于像素算术的编码空间 [25, 26]。Zhu等将SHA-3明文哈希与并行DNA编解码结合，在保持安全性的同时降低时间开销 [27]；Cun等采用动态DNA编码与RNA计算，并配合置换序列实施行列循环移位 [28]；Afify等指出，对整幅图像施加单一、易检测的DNA编码规则几乎不能提高安全性，因而转向动态多规则编码 [29]。Yu等进一步以DNA网络与超混沌系统构造加密算法 [30]。近两年，DNA机制继续向三维结构、密钥校验与扩散层延伸：Lai等将超混沌映射与DNA立方体用于医疗物联网图像加密 [31]；Su等利用DNA三链结构完成置乱与运算 [32]；Al-dayel等以最大Lyapunov指数校验的混沌密钥流驱动DNA扩散 [33]；Huang等将位平面分解、DNA编码与五维哈密顿超混沌相结合 [26]；Li等提出双混沌系统与动态DNA编码方案，以兼顾安全性与计算效率 [34]；Mei等采用余弦调制混沌与动态DNA编码 [35]；Suzgen等则把忆阻混沌与DNA-ARX-3DES混合用于医学图像加密 [36]。这些工作表明，动态规则、并行编解码和明文关联密钥已能改善DNA加密的扩散性能与抗选择明文能力；但其编码单元多在整幅图像或固定分块上均匀施加，规则选择与图像局部纹理、通道结构的适配仍然有限，也较少与时空混沌格子的格点序列形成块级对应。因此，如何按明文结构自适应划分DNA编码块、并使之与CML密钥流协同，仍有待解决。

Fridrich提出的扰乱与扩散两步结构至今仍是混沌图像加密的主流框架：扰乱打乱像素或比特的空间位置，扩散则使像素值的局部改变扩展到密文全局 [1, 37]。随着对该框架的分析不断深入，仅依赖可逆置换或弱扩散、且两阶段可分离的方案已暴露出安全隐患。Li表明，层次化置换类算法可被针对性破解 [38]。Arnold映射、规则行列置换以及标准Zigzag扫描因轨道周期或固定路径而易于被统计分析利用；后续工作虽以动态行置乱、滑动Zigzag或魔方置乱等加以替换 [39–41]，但两阶段串联的总体结构并未改变。比特级扰乱将像素拆为二进制后再操作，使一次置换同时改变位置与灰度，安全性通常高于像素级置换 [42, 43]。近两年，这一路线被进一步细化：Lu等给出比特级多阶段动态扰乱与超混沌映射相结合的加密范式 [44]；Li等在三维超混沌驱动下引入位平面内部混淆与全局置乱扩散 [45]；Yan等将比特级置换与双向列级联扩散嵌入高维神经混沌系统 [46]；Wang等则尝试由元胞自动机控制同步置乱–扩散，以削弱两步可分性 [23]。尽管如此，多数方案仍将扰乱与扩散作为可独立逆向的两个模块，扩散路径也多为固定扫描或单向密文反馈，对跨通道、跨位平面的联合隐藏以及局部–全局雪崩的同时强化仍然不足。这为在DNA张量上实施三维联合置乱，并以块级运算与全局动态路径密文反馈构成两级扩散，留下了明确的改进空间。

为解决上述问题，本文的主要研究贡献如下：

(1) 本文提出S-NCML（Salomon Non-adjacent Coupled Map Lattice），一种以Salomon映射为局部动力学、同时引入近邻余弦耦合与非邻耦合的时空混沌系统。该系统在广泛的参数区间内保持强混沌，能够提供更大的密钥空间，其复杂的伪随机序列增强了后续加密算法的安全性。

(2) 本文提出一种明文关联的自适应分块DNA彩色图像加密框架。利用明文哈希与用户密钥共同驱动混沌参数与格点初值，并按图像梯度将彩色图像递归划分为不同尺寸的矩形块；每个块采用独立的DNA编码规则，将24个RGB位平面编码为12个DNA平面，从而同时适应不同通道、不同尺寸与不同纹理复杂度的图像。

(3) 本文提出DNA三维联合置乱：在块内沿DNA平面、行、列三轴独立置换，并在全局对DNA张量实施由混沌矩阵驱动的三通道循环移位。该操作同时打乱空间位置与跨通道位信息，从而隐藏明文的结构与色彩特征。

(4) 本文提出一种两级DNA扩散机制，具体为块级动态DNA运算与全局动态路径密文反馈扩散。后者沿混沌置换路径将像素分组，组间以密文向量作为前驱进行耦合，从而增强局部与全局的雪崩效应，提升抗差分攻击与选择明文攻击的能力。

# 2 加密与解密算法

## 2.1 密钥设计与混沌序列生成

为提升密码系统的抗攻击性能，抵御选择明文攻击（CPA）与已知明文攻击（KPA），本方案设计了一种基于明文图像与外部密钥双驱动的参数及密钥流生成机制。系统利用尺寸为 $M \times N$ 的明文图像 $P$ 通过 SHA-384 哈希函数生成 384 位内部密钥 $K$，并与通信双方预先商定的 384 位外部密钥 $E$ 共同作用。将 $K$ 与 $E$ 分别均分为 8 个 48 位的子块：$K = [k_1, k_2, \ldots, k_8]$ 与 $E = [e_1, e_2, \ldots, e_8]$，其中每个子块的取值范围为 $[0, 2^{48} - 1]$。

为消除各子块间的局部相关性，系统采用双向滑动窗口结构进行交叉扰动：当滑动窗口在内部密钥 $K$ 上正向滑动时，在外部密钥 $E$ 上逆向滑动。五个中间归一化变量 $V_i \in [0, 1)$（$i = 1, 2, \ldots, 5$）的计算公式如下：

$$\begin{cases} V_1 = \dfrac{(k_1 \oplus k_2 \oplus k_3 \oplus k_4) + (e_8 \oplus e_7 \oplus e_6 \oplus e_5)}{2 \times 2^{48}}, \\ V_2 = \dfrac{(k_2 \oplus k_3 \oplus k_4 \oplus k_5) + (e_7 \oplus e_6 \oplus e_5 \oplus e_4)}{2 \times 2^{48}}, \\ V_3 = \dfrac{(k_3 \oplus k_4 \oplus k_5 \oplus k_6) + (e_6 \oplus e_5 \oplus e_4 \oplus e_3)}{2 \times 2^{48}}, \\ V_4 = \dfrac{(k_4 \oplus k_5 \oplus k_6 \oplus k_7) + (e_5 \oplus e_4 \oplus e_3 \oplus e_2)}{2 \times 2^{48}}, \\ V_5 = \dfrac{(k_5 \oplus k_6 \oplus k_7 \oplus k_8) + (e_4 \oplus e_3 \oplus e_2 \oplus e_1)}{2 \times 2^{48}}, \end{cases} \tag{1}$$

式中 $\oplus$ 表示按位异或运算。由于每个 4 块异或项的最大取值为 $2^{48} - 1$，分子的算术加和严格小于 $2 \times 2^{48}$，因此式 (1)保证了各 $V_i \in [0, 1)$。随后，利用正弦非线性折叠消除线性关系，提取出系统的动力学控制参数 $(\mu, v, \alpha, \beta)$ 及混沌初值 $x_0$：

$$\begin{cases} \mu = \mathrm{mod}\left(\left\lfloor |\sin(4\pi \cdot V_1)| \times 10^6 \right\rfloor, 10\right), \\ v = \mathrm{mod}\left(\left\lfloor |\sin(4\pi \cdot V_2)| \times 10^6 \right\rfloor, 10\right), \\ \alpha = \mathrm{mod}\left(\left\lfloor |\sin(4\pi \cdot V_3)| \times 10^6 \right\rfloor, 10\right), \\ \beta = \mathrm{mod}\left(\left\lfloor |\sin(4\pi \cdot V_4)| \times 10^6 \right\rfloor, 10\right), \\ x_0 = \mathrm{mod}\left(|\sin(4\pi \cdot V_5)| \times 10^6, 1\right), \end{cases} \tag{2}$$

其中 $\lfloor \cdot \rfloor$ 为向下取整函数，$\mathrm{mod}(z, 1) = z - \lfloor z \rfloor$ 表示取小数部分，使得初值 $x_0 \in (0, 1)$。将派生得到的初值 $x_0$ 和参数输入离散混沌系统：

$$x_{t+1} = \left|\sin\left((5 + 3\mu)\left(1 - v \cdot x_t \cdot \sin\left(15\pi x_t(1 - x_t)\right)\right)\right)\right|. \tag{3}$$

为消除初始暂态效应，系统迭代前 $N_0$ 次被舍弃，继续收集后续 $L = M \times N$ 次迭代输出，构成最终的伪随机密钥流序列 $X = \{x_1, x_2, \ldots, x_L\}$。该算法的完整执行过程如算法 1 所示。

**Algorithm 1** Parameters and Keystream Generation

**Require:** Plaintext image $P$ of size $M \times N$, 384-bit external secret key $E$, transient iteration count $N_0$, keystream length $L$.

**Ensure:** Chaotic parameters $(\mu, v, \alpha, \beta)$, initial state $x_0$, pseudorandom keystream $X$.

$H \leftarrow \text{SHA-384}(P)$;
Split $H$ into 8 blocks of 48-bit integers: $K = [k_1, k_2, \ldots, k_8]$;
Split $E$ into 8 blocks of 48-bit integers: $E = [e_1, e_2, \ldots, e_8]$;
**for** $i = 1$ **to** 5 **do**
    $S_K \leftarrow k_i \oplus k_{i+1} \oplus k_{i+2} \oplus k_{i+3}$;
    $S_E \leftarrow e_{9-i} \oplus e_{8-i} \oplus e_{7-i} \oplus e_{6-i}$;
    $V_i \leftarrow \frac{S_K + S_E}{2 \times 2^{48}}$;
**end for**
$\mu \leftarrow \text{mod}\left(\lfloor |\sin(4\pi \cdot V_1)| \times 10^6 \rfloor, 10\right)$;
$v \leftarrow \text{mod}\left(\lfloor |\sin(4\pi \cdot V_2)| \times 10^6 \rfloor, 10\right)$;
$\alpha \leftarrow \text{mod}\left(\lfloor |\sin(4\pi \cdot V_3)| \times 10^6 \rfloor, 10\right)$;
$\beta \leftarrow \text{mod}\left(\lfloor |\sin(4\pi \cdot V_4)| \times 10^6 \rfloor, 10\right)$;
$x_0 \leftarrow \text{mod}\left(|\sin(4\pi \cdot V_5)| \times 10^6, 1\right)$;
$x \leftarrow x_0$;
**for** $t = 1$ **to** $N_0 + L$ **do**
    $x \leftarrow |\sin\left((5 + 3\mu)\left(1 - v \cdot x \cdot \sin(15\pi x(1 - x))\right)\right)|$;
    **if** $t > N_0$ **then**
        $X[t - N_0] \leftarrow x$;
    **end if**
**end for**
**return** $\mu, v, \alpha, \beta, x_0, X$;

# 3 性能分析

该算法的优缺点需要通过仿真和安全测试验证。因此,本部分介绍算法的仿真及性能测试结果。仿真实验在AMD Ryzen 7 4800H CPU,16 GB 内存,Windows 11的，MATLAB 2024b环境下进行，对多张尺寸不同的图片进行了仿真测试。

## 3.1 仿真结果

从USC-SIPI图像数据库中选择了3张$256 \times 256$、一张$512 \times 512$和一张$128 \times 128$的图像进行仿真测试。图[fig]展示了采用所提加密算法得到的加密图像和解密图像。可以看出，加密图像为不规则噪声信号图像，难以从中获取原始图像的任何信息；解密图像则与原始图像一致。

## 3.2 秘钥空间分析

所有可能秘钥的总数称为秘钥空间。秘钥空间越大，算法的安全性越高。Alvarez[ref]指出,安全的加密算法的秘钥空间至少应为$2^{100}$，且加密算法的秘钥空间越大，其抵抗暴力破解的能力就越强。本文所提加密算法可以是384位哈希值H，或是五个参数$\{\mu, v, \alpha, \beta, A_0\}$。(1)若秘钥为384位哈希值H，则秘钥空间为$2^{384}$；(2)若秘钥为五个参数$\{\mu, v, \alpha, \beta, A_0\}$，在计算机精度为$10^{-15}$的情况下，则秘钥空间为$10^{5\times15} = 10^{75} > 2^{249}$。考虑到攻击者可能通过暴力破解或其他方式获取秘钥，最终秘钥空间应为两者的最小值，即$10^{75} > 2^{249}$，因此本文所提加密算法的秘钥空间足够大，能够抵抗暴力破解攻击。

## 3.3 直方图分析

# 参考文献


[1] Guanrong Chen, Yaobin Mao, and Charles K. Chui. A symmetric image encryption scheme based on 3D chaotic cat maps. *Chaos, Solitons & Fractals*, 21(3):749–761, 2004.

[2] Morteza SaberiKamarposhti, Amirabbas Ghorbani, and Mehdi Yadollahi. A comprehensive survey on image encryption: Taxonomy, challenges, and future directions. *Chaos, Solitons & Fractals*, 178:114361, 2024.

[3] Xingyuan Wang, Nana Guan, Hongyu Zhao, Siwei Wang, and Yingqian Zhang. A new image encryption scheme based on coupling map lattices with mixed multi-chaos. *Scientific Reports*, 10(1):9784, 2020.

[4] Kunihiko Kaneko. Spatiotemporal chaos in one- and two-dimensional coupled map lattices. *Physica D: Nonlinear Phenomena*, 37(1-3):60–82, 1989.

[5] Farhad Khellat, Akashe Ghaderi, and Nastaran Vasegh. Li–yorke chaos and synchronous chaos in a globally nonlocal coupled map lattice. *Chaos, Solitons & Fractals*, 44(11):934–939, 2011.

[6] Rhouma Rhouma, Soumaya Meherzi, and Safya Belghith. Ocml-based colour image encryption. *Chaos, Solitons & Fractals*, 40(1):309–318, 2009.

[7] Xingyuan Wang and Pengbo Liu. A new full chaos coupled mapping lattice and its application in privacy image encryption. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 69(3):1291–1301, 2022.

[8] Ying-Qian Zhang and Xing-Yuan Wang. Spatiotemporal chaos in mixed linear–nonlinear coupled logistic map lattice. *Physica A: Statistical Mechanics and its Applications*, 402:104–118, 2014.

[9] Ying-Qian Zhang and Xing-Yuan Wang. A symmetric image encryption algorithm based on mixed linear–nonlinear coupled map lattice. *Information Sciences*, 273:329–351, 2014.

[10] Ying-Qian Zhang and Xing-Yuan Wang. A new image encryption algorithm based on non-adjacent coupled map lattices. *Applied Soft Computing*, 26:10–20, 2015.

[11] Xingyuan Wang, Le Feng, Rui Li, and Fuchen Zhang. A fast image encryption algorithm based on non-adjacent dynamically coupled map lattice model. *Nonlinear Dynamics*, 95(4):2797–2824, 2019.

[12] Xingyuan Wang, Hongyu Zhao, and Mingxu Wang. A new image encryption algorithm with nonlinear-diffusion based on multiple coupled map lattices. *Optics & Laser Technology*, 115:42–57, 2019.

[13] Ye Tao, Wenhua Cui, and Zhao Zhang. Spatiotemporal chaos in multiple dynamically coupled map lattices and its application in a novel image encryption algorithm. *Journal of Information Security and Applications*, 55:102650, 2020.

[14] Xingyuan Wang, Le Feng, Shibing Wang, Chuan Zhang, and Yingqian Zhang. Spatiotemporal chaos in coupled logistic map lattice with dynamic coupling coefficient and its application in image encryption. *IEEE Access*, 6:39705–39724, 2018.

[15] Mingxu Wang, Xingyuan Wang, Chunpeng Wang, Zhiqiu Xia, Hongyu Zhao, Suo Gao, Shuang Zhou, and Nianmin Yao. Spatiotemporal chaos in cross coupled map lattice with dynamic coupling coefficient and its application in bit-level color image encryption. *Chaos, Solitons & Fractals*, 139:110028, 2020.

[16] Xingyuan Wang and Jingjing Yang. Spatiotemporal chaos in multiple coupled mapping lattices with multi-dynamic coupling coefficient and its application in color image encryption. *Chaos, Solitons & Fractals*, 147:110970, 2021.

[17] Xingyuan Wang and Jingjing Yang. A privacy image encryption algorithm based on piecewise coupled map lattice with multi dynamic coupling coefficient. *Information Sciences*, 569:217–240, 2021.

[18] Ying-Qian Zhang, Yi He, and Xing-Yuan Wang. Spatiotemporal chaos in mixed linear–nonlinear two-dimensional coupled logistic map lattice. *Physica A: Statistical Mechanics and its Applications*, 490:148–160, 2018.

[19] Longbiao Du, Lin Teng, Huipeng Liu, and Hanrui Lu. Multiple face images encryption based on a new non-adjacent dynamic coupled mapping lattice. *Expert Systems with Applications*, 238:121728, 2024.

[20] Wei Li, Lingling Lu, Yang Yu, Chengye Zou, and Yunong Liu. Multiple face encryption based on non-adjacent coupled map lattice and improved yolov8 algorithm. *Journal of King Saud University Computer and Information Sciences*, 37(10):321, 2025.

[21] Dawei Ding, Tao Liu, Hong Cheng, Zongli Yang, Xiang Liu, and Haitao Zhou. Spatiotemporal chaos in exponential cosine polynomial nonlinear coupling for image encryption: an adaptive multi-image compression encryption scheme with autoencoder. *Nonlinear Dynamics*, 114(5):353, 2026.

[22] Jiakun Wang, Zhen Li, Weijie Tan, and Xianming Wu. Plaintext-related image encryption scheme based on a new sinusoidal-quadratic map CML and random-trajectory josephus permutation. *Physica Scripta*, 101(1):015213, 2026.

[23] Jiakun Wang, Zhen Li, Weijie Tan, and Xianming Wu. A novel image encryption scheme with synchronous permutation–diffusion controlled by CA and a sine-cosine-optimized CML model. *The European Physical Journal Plus*, 140(9):841, 2025.

[24] Zhuo Liu, Yong Wang, Jinyuan Liu, Jun Feng, and Leo Yu Zhang. A newly image encryption scheme based on 3-D coupled map lattice and baker map. *Cybersecurity*, 9(1):103, 2026.

[25] Shaohui Yan, Lin Li, Binxian Gu, Xi Sun, Yu Ren, and Yuyan Zhang. A color image encryption scheme based on chaotic mapping, chaotic system, and DNA coding. *Applied Intelligence*, 53(24):31181–31206, 2023.

[26] Li Huang, Cong Ding, Zhenjie Bao, Haitao Chen, and Changsheng Wan. A DNA encoding image encryption algorithm based on chaos. *Mathematics*, 13(8):1330, 2025.

[27] Shenli Zhu, Xiaoheng Deng, Wendong Zhang, and Congxu Zhu. Image encryption scheme based on newly designed chaotic map and parallel dna coding. *Mathematics*, 11(1), 2023.

[28] Qiqi Cun, Xiaojun Tong, Zhu Wang, and Miao Zhang. A new chaotic image encryption algorithm based on dynamic DNA coding and RNA computing. *The Visual Computer*, 39(12):6589–6608, 2023.

[29] Yasmine M. Afify, Nada H. Sharkawy, Walaa Gad, and Nagwa Badr. A new dynamic DNA-coding model for gray-scale image encryption. *Complex & Intelligent Systems*, 10(1):745–761, 2024.

[30] Jinwei Yu, Kaiyu Peng, Langwen Zhang, and Wei Xie. Image encryption algorithm based on DNA network and hyperchaotic system. *The Visual Computer*, 40(11):8001–8021, 2024.

[31] Qiang Lai and Hanqiang Hua. Secure medical image encryption scheme for healthcare IoT using novel hyperchaotic map and DNA cubes. *Expert Systems with Applications*, 264:125854, 2025.

[32] Yingjie Su, Han Xia, Ziyu Chen, Han Chen, and Linqing Huang. A novel one-dimensional chaotic system for image encryption through the three-strand structure of DNA. *Entropy*, 27(8):776, 2025.

[33] Ibrahim Al-dayel, Muhammad Faisal Nadeem, Yasir Bashir, and Ayesha Shabbir. Information-theoretic medical image encryption via LLE-verified chaotic keystreams and DNA diffusion. *Entropy*, 27(11):1149, 2025.

[34] Rui Li, Zefeng Zhang, Yuan Zhu, Chuanhong Du, and Licai Liu. Image encryption algorithm based on dual chaotic systems and dynamic DNA encoding. *Physica Scripta*, 101(25):255210, 2026.

[35] Junhua Mei, Bingchan Fan, and Shan Wu. A novel image encryption algorithm based on cosine-modulated chaotic system and dynamic DNA coding. *Discover Applied Sciences*, 2026.

[36] Enes Eren Suzgen, Muhammet Emin Sahin, and Hasan Ulutas. A novel hybrid medical image encryption scheme based on memristive chaos and DNA-ARX-3DES with real-time implementation. *Scientific Reports*, 16(1):6230, 2026.

[37] Jiri Fridrich. Symmetric ciphers based on two-dimensional chaotic maps. *International Journal of Bifurcation and Chaos*, 08(06):1259–1284, 1998.

[38] Chengqing Li. Cracking a hierarchical chaotic image encryption algorithm based on permutation. *Signal Processing*, 118:203–210, 2016.

[39] Xingyuan Wang and Xuan Chen. An image encryption algorithm based on dynamic row scrambling and zigzag transformation. *Chaos, Solitons & Fractals*, 147:110962, 2021.

[40] Zhixuan Yang and Lin Teng. A color image encryption algorithm using slide zigzag scrambling with two-dimensional hyperchaotic system. *Physica Scripta*, 100(6):065220, 2025.

[41] S. Sharmila, R.S. Bhuvaneswaran, and Dhandapani Vaithiyanathan. Secure image encryption using rubik's cube-based scrambling with chaos-driven diffusion and circular shifts. *Optik*, 339:172533, 2025.

[42] Zhi-liang Zhu, Wei Zhang, Kwok-wo Wong, and Hai Yu. A chaos-based symmetric image encryption scheme using a bit-level permutation. *Information Sciences*, 181(6):1171–1186, 2011.

[43] Lu Xu, Zhi Li, Jian Li, and Wei Hua. A novel bit-level image encryption algorithm based on chaotic maps. *Optics and Lasers in Engineering*, 78:17–25, 2016.

[44] Jun Lu, Kuoyu Wu, Jiaxin Zhang, and Xue Xia. An image encryption paradigm: Combining bit-level multi-stage dynamic scrambling with a hyperchaotic map. *Digital Signal Processing*, 168:105488, 2026.

[45] Weifeng Li, Bin Ge, Chenxing Xia, Menglin Ma, and Xiyun Huang. Image encryption algorithm for 3D-hyperchaotic systems driven by nonlinear feedback mechanism. *Signal, Image and Video Processing*, 19(12):984, 2025.

[46] Lin Yan, Sheng Dong, and Weiqing Liu. Efficient image encryption scheme based on 7D-FHN neural network with bit-level permutation and bidirectional column cascaded diffusion. *Physica Scripta*, 101(17):175204, 2026.